Linux File System Explorer

/var

```
user@webterm:~$ ls /var
log/
user@webterm:~$ ls /var/log
batch/  journal/  system.log
```

/home

```
user@webterm:~$ ls /home
user/
```

/etc

```
user@webterm:~$ ls /etc
hosts  passwd
```

/tmp

```
user@webterm:~$ ls -la /tmp
total 8
drwxrwxrwt 2 root root 4096 Oct  9 22:43 ./
drwxr-xr-x 7 root root 4096 Oct  9 22:43 ../
```

/usr

```
user@webterm:~$ ls /usr
bin/  share/
user@webterm:~$ ls /usr/bin
alias  awk  bg  bunzip2  bzip2  cat  cd  chgrp  chmod  chown  clear  cp  crontab  curl
date  df  du  echo  fg  file  find  grep  gunzip  gzip  head  help  history  jobs  kill
less  ln  locate  ls  man  mkdir  more  mount  mv  ps  pwd  rm  rmdir  rsync  scp  sed
sort  ssh  stat  systemctl  tail  tar  tee  top  touch  umask  umount  uniq  unzip
watch  wc  webterm  whereis  which  xargs  zip
```

Linux Commands Investigation

`Pwd`

```
user@webterm:~$ pwd
/home/user
```

`cd`

```
user@webterm:~$ cd /tmp
user@webterm:/tmp$ pwd
/tmp
```

`ls` and `ls -la`

```
user@webterm:~$ ls
documents/  projects/
user@webterm:~$ ls -la
total 24
drwxr-xr-x 4 user user 4096 Oct  9 22:43 ./
drwxr-xr-x 3 root root 4096 Oct  9 22:43 ../
-rw-r--r-- 1 user user   46 Oct  9 22:43 .gitconfig
-rw-r--r-- 1 user user  113 Oct  9 22:43 .zshrc
drwxr-xr-x 2 user user 4096 Oct  9 22:43 documents/
drwxr-xr-x 2 user user 4096 Oct  9 22:43 projects/
```

`Mkdir`

```
user@webterm:/$ cd /tmp
user@webterm:/tmp$ pwd
/tmp
user@webterm:/tmp$ mkdir directory1
user@webterm:/tmp$ ls -ld directory1
drwxr-xr-x 2 user user 4096 Oct  9 23:11 directory1/
```

rm

```
user@webterm:/tmp$ touch testfile.txt
user@webterm:/tmp$ ls
directory1/  testfile.txt
user@webterm:/tmp$ rm testfile.txt
user@webterm:/tmp$ ls
directory1/
user@webterm:/tmp$ ls -l testfile.txt
```

Linux Permissions

ls -l

```
user@webterm:~/documents$ ls -l
total 8
-rw-r--r-- 1 user user 143 Oct  9 22:20 readme.txt
-rw-r--r-- 1 user user 105 Oct  9 22:20 sample.txt
user@webterm:~/documents$
```

ls -l readme.txt

```
user@webterm:~/documents$ ls -l readme.txt
-rw-r--r-- 1 user user 143 Oct  9 22:20 readme.txt
user@webterm:~/documents$
```

chmod 755

```
user@webterm:~/documents$ ls -l readme.txt
-rw-r--r-- 1 user user 143 Oct  9 22:20 readme.txt
user@webterm:~/documents$ chmod 755 readme.txt
user@webterm:~/documents$ ls -l readme.txt
-rwxr-xr-x 1 user user 143 Oct  9 22:20 readme.txt
user@webterm:~/documents$
```

`chmod 644`

```
user@webterm:~/documents$ ls -l sample.txt
-rwxr-xr-x 1 user user 105 Oct  9 22:20 sample.txt
user@webterm:~/documents$ chmod 644 sample.txt
user@webterm:~/documents$ ls -l sample.txt
-rw-r--r-- 1 user user 105 Oct  9 22:20 sample.txt
user@webterm:~/documents$
```

`chmod 777`

```
user@webterm:~/documents$ ls -l readme.txt
-rwxr-xr-x 1 user user 143 Oct  9 22:20 readme.txt
user@webterm:~/documents$ chmod 777 readme.txt
user@webterm:~/documents$ ls -l readme.txt
-rwxrwxrwx 1 user user 143 Oct  9 22:20 readme.txt
user@webterm:~/documents$
```

Linux Processes

- What Is a Linux Process?

```
user@webterm:~/documents$ sleep 100 &
[1] 10000
user@webterm:~/documents$ ps
  PID TTY          TIME CMD
 1234 pts/0    00:00:00 zsh
 2847 pts/0    00:01:23 node
 3456 pts/0    02:34:56 python3
 2345 pts/0    00:00:00 ps
10000 pts/0    00:00:00 sleep
user@webterm:~/documents$
```

- What Does `ps` Show?

```
user@webterm:~/documents$ ps
  PID TTY          TIME CMD
 1234 pts/0    00:00:00 zsh
 2847 pts/0    00:01:23 node
 3456 pts/0    02:34:56 python3
 2345 pts/0    00:00:00 ps
user@webterm:~/documents$
```

- What Does `ps aux` Show?

```
user@webterm:~/documents$ ps aux
USER         PID %CPU %MEM    VSZ   RSS TTY      STAT START   TIME COMMAND
root           1  0.0  0.1 169232 13120 ?        Ss   00:00   0:02 /sbin/init
root           2  0.0  0.0      0     0 ?        S    00:00   0:00 [kthreadd]
root         123  0.3  0.2 120344  9876 ?        Ss   00:01   0:01 /lib/systemd/systemd-journald
root         456  0.0  0.1  45672  5432 ?        Ss   00:01   0:00 /usr/sbin/sshd -D
user         789  0.1  0.1  45672  5432 ?        S    10:30   0:00 sshd: user@pts/0
user        1234  0.2  0.1  23456  3456 pts/0    Ss   10:30   0:00 -zsh
user        2847  1.2  1.5 345678 56789 pts/0    S    10:35   1:23 node server.js
user        3456 98.5  4.2 456789 87654 pts/0    R    09:15 154:56 python3 backup_sync.py
root        5678  0.0  0.0  12345  2345 ?        Ss   00:01   0:00 /usr/sbin/crond
root         301  0.0  0.0   5678  1234 ?        Ss   00:00   0:00 /etc/init.d/networking
systemd+     201  0.0  0.1  24576  6789 ?        Ss   00:00   0:01 /lib/systemd/systemd-resolved
user        2345  0.0  0.0  11234  1234 pts/0    R+   10:45   0:00 ps
user@webterm:~/documents$
```

- What Does `top` Do?

```
user@webterm:~/documents$ top -b -n 1
top - 23:13:36 up 10:30:45, 2 users,  load average: 0.12, 0.15, 0.10
Tasks: 12 total,   2 running, 10 sleeping,   0 stopped,   0 zombie
%Cpu(s):  3.0 us,  1.3 sy,  0.0 ni, 95.4 id,  0.3 wa,  0.0 hi,  0.0 si,  0.0 st
MiB Mem :   7948.0 total,   1843.0 free,   3277.0 used,   2828.0 buff/cache
MiB Swap:   2048.0 total,   1920.0 free,   128.0 used.   4200.0 avail Mem

  PID USER      PR  NI    VIRT    RES    SHR S  %CPU  %MEM     TIME+ COMMAND
 3456 user      20   0  446.1m   85.6m   59.9m R  98.5   4.2   2:34:56 python3
 2345 user      20   0   11.0m    1.2m    987k R   1.5   0.0   0:00.01 top
 2847 user      20   0  337.6m   55.5m   38.8m S   1.2   1.5      1:23 node
  123 root      20   0  117.5m    9.6m    6.8m S   0.3   0.2      0:01 systemd-j
 1234 user      20   0   22.9m    3.4m    2.4m S   0.2   0.1      0:00 zsh
  789 user      20   0   44.6m    5.3m    3.7m S   0.1   0.1      0:00 sshd
    1 root      20   0  165.3m   12.8m    9.0m S   0.0   0.1      0:02 systemd
    2 root      20   0       0k      0k      0k S   0.0   0.0      0:00 kthreadd
  456 root      20   0   44.6m    5.3m    3.7m S   0.0   0.1      0:00 sshd
 5678 root      20   0   12.1m    2.3m    1.6m S   0.0   0.0      0:00 crond
  301 root      20   0    5.5m    1.2m    863k S   0.0   0.0      0:00 networking
  201 systemd-  20   0   24.0m    6.6m    4.6m S   0.0   0.1      0:01 systemd-r
user@webterm:~/documents$
```

- Why Would a Developer Investigate Processes?

```
user@webterm:~/documents$ ps aux
USER         PID %CPU %MEM    VSZ   RSS TTY      STAT START    TIME COMMAND
root           1  0.0  0.1 169232 13120 ?        Ss   00:00    0:02 /sbin/init
root           2  0.0  0.0      0     0 ?        S    00:00    0:00 [kthreadd]
root         123  0.3  0.2 120344  9876 ?        Ss   00:01    0:01 /lib/systemd/systemd-journald
root         456  0.0  0.1  45672  5432 ?        Ss   00:01    0:00 /usr/sbin/sshd -D
user         789  0.1  0.1  45672  5432 ?        S    10:30    0:00 sshd: user@pts/0
user        1234  0.2  0.1  23456  3456 pts/0    Ss   10:30    0:00 -zsh
user        2847  1.2  1.5 345678 56789 pts/0    S    10:35    1:23 node server.js
user        3456 98.5  4.2 456789 87654 pts/0    R    09:15  154:56 python3 backup_sync.py
root        5678  0.0  0.0  12345  2345 ?        Ss   00:01    0:00 /usr/sbin/crond
root         301  0.0  0.0   5678  1234 ?        Ss   00:00    0:00 /etc/init.d/networking
systemd+     201  0.0  0.1  24576  6789 ?        Ss   00:00    0:01 /lib/systemd/systemd-resolved
user        2345  0.0  0.0  11234  1234 pts/0    R+   10:45    0:00 ps
user       10000  0.0  0.1  12345  2345 pts/0    S    10:45    0:00 sleep 100
user       10001  0.0  0.1  12345  2345 pts/0    S    10:45    0:00 sleep 100
user       10002  0.0  0.1  12345  2345 pts/0    S    10:45    0:00 sleep 100
user       10003  0.0  0.1  12345  2345 pts/0    S    10:45    0:00 sleep 100
user@webterm:~/documents$ top -b -n 1
top - 23:19:58 up 10:30:45, 2 users,  load average: 0.12, 0.15, 0.10
Tasks: 12 total,   2 running, 10 sleeping,   0 stopped,   0 zombie
%Cpu(s):  3.0 us,  1.3 sy,  0.0 ni, 95.4 id,  0.3 wa,  0.0 hi,  0.0 si,  0.0 st
MiB Mem :   7948.0 total,   1843.0 free,   3277.0 used,   2828.0 buff/cache
MiB Swap:   2048.0 total,   1920.0 free,   128.0 used.   4200.0 avail Mem

  PID USER      PR  NI    VIRT    RES    SHR S  %CPU  %MEM     TIME+ COMMAND
 3456 user      20   0  446.1m   85.6m   59.9m R  98.5   4.2   2:34:56 python3
 2345 user      20   0   11.0m    1.2m    987k R   1.5   0.0   0:00.01 top
 2847 user      20   0  337.6m   55.5m   38.8m S   1.2   1.5      1:23 node
  123 root      20   0  117.5m    9.6m    6.8m S   0.3   0.2      0:01 systemd-j
 1234 user      20   0   22.9m    3.4m    2.4m S   0.2   0.1      0:00 zsh
  789 user      20   0   44.6m    5.3m    3.7m S   0.1   0.1      0:00 sshd
    1 root      20   0  165.3m   12.8m    9.0m S   0.0   0.1      0:02 systemd
    2 root      20   0      0k      0k      0k S   0.0   0.0      0:00 kthreadd
  456 root      20   0   44.6m    5.3m    3.7m S   0.0   0.1      0:00 sshd
 5678 root      20   0   12.1m    2.3m    1.6m S   0.0   0.0      0:00 crond
  301 root      20   0    5.5m    1.2m    863k S   0.0   0.0      0:00 networking
  201 systemd-  20   0   24.0m    6.6m    4.6m S   0.0   0.1      0:01 systemd-r
user@webterm:~/documents$
```